

01_Writing_Simple_Function

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Let's make a first function. for example, a function that returns the double of a number

```
[ ]: x=5

#what we want
result=x*2
print(result)
```

```
[ ]: def myfunc(x):
      result=x*2
      return result
```

```
[ ]: #now we can use it
myfunc(5)
```

```
[ ]:
```

```
[ ]: y=5
myfunc(y)
```

The name of the variable i pass does not need to be the same as the one in the definition.

I can make a function without argument

```
[1]: def sayHello():
      print('Hello')
```

```
[2]: sayHello()
```

Hello

```
[3]: # Celsius to Farenheit converter
# formula is  $(x^{\circ}C \times 9/5) + 32 = X^{\circ}F$ 

def C2F(x):
    F=x*9/5+32
    return F

x=23
print( 'A temperature of',x,'celsius is equal to',C2F(x),'Farenheit')
```

A temperature of 23 celsius is equal to 73.4 Farenheit

```
[4]: # example. write a function that gets a list of characters and remove all the vowels
```

```
def removevowels(l):
    newlist=[c for c in l if c not in 'aeiouAEIOU']
    return newlist

l1=['o', 'z', 'i', 'r', 'p']
removevowels(l1)
```

```
[4]: ['z', 'r', 'p']
```

```
[ ]:
```

Rescale a numpy array

```
[14]: import numpy as np
a=np.array( [ [3,-1,5], [0,7,3]])
a
```

```
[14]: array([[ 3, -1,  5],
           [ 0,  7,  3]])
```

we want to make the transformation $(a-\min(a))/(\max(a)-\min(a))$ That is, we normalize the array to be between 0 and 1 $\frac{a-\min(a)}{\max(a)-\min(a)}$

```
[17]: def normalise(a):
        centered_a=(a-np.min(a))
        normal_a=centered_a/np.max(centered_a)
        return normal_a

normalise(a)
```

```
[17]: array([[0.5 , 0.   , 0.75 ],
           [0.125, 1.   , 0.5  ]])
```

```
[18]: normalise(3*a+5)
```

```
[18]: array([[0.5 , 0.   , 0.75 ],
           [0.125, 1.   , 0.5  ]])
```

```
[ ]:
```